<InfPALS/>

Early Careers Planning

Skills Week 7

Wednesday 14:10

Hanson

BSc Artificial Intelligence and Computer Science - Year 2

Kenton

MInf Informatics - Year 2

Aman

BEng Computer Science - Year 2

Agenda

- Introduction
- Self-assessment
- Career options in informatics
- Choosing future university courses

Introductions

Importance of early career planning

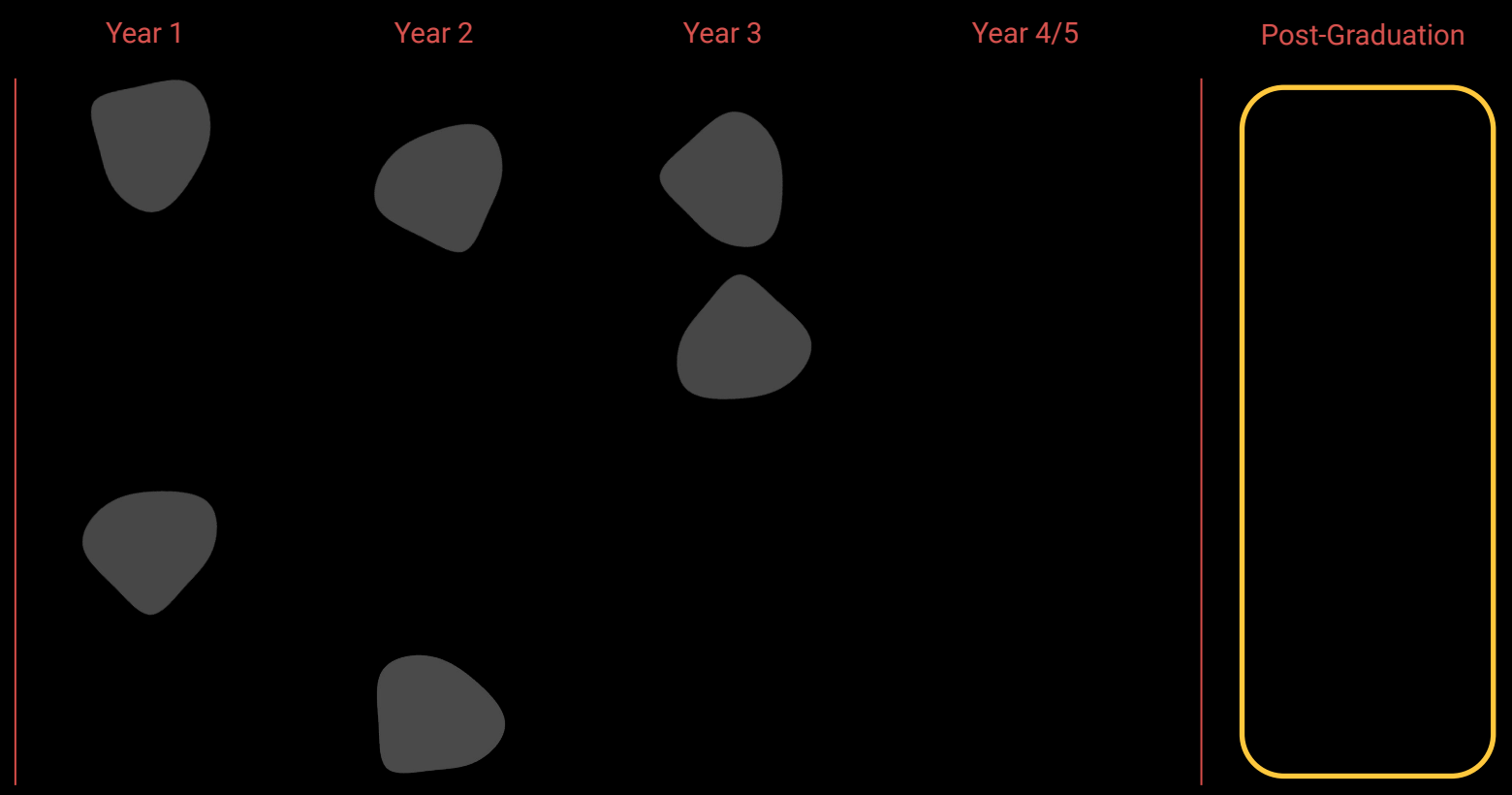
- Computer science is a broad field with many different career paths.
  - By understanding your career goals, you can choose courses that will help you develop the skills and knowledge you need to be successful.
  - For example, if you're interested in a career in software engineering, you'll want to focus on courses that teach you software design, algorithms, etc.
- The job market for computer science graduates is very competitive.
- Students who have a clear plan for their careers are more likely to achieve their goals.
  - This does not mean you have to follow one path, but by making goals you can make purposeful changes to your career.

Setting Career Goals

> ### What are career goals?
- Career goals are the specific things that you want to achieve in your professional life.
- They can be short-term or long-term, and they can be related to your education, your job, or your overall career path.

> ### SMART goals can help make targets more concrete:
- Specific: Your goals should be specific and well-defined.
- Measurable: Your goals should be measurable so that you can track your progress and see how close you are to achieving them.
- Achievable: Your goals should be achievable but challenging.
- Relevant: Your goals should be relevant to your overall goals and objectives.
- Time-bound: Your goals should have a specific deadline.

Career Plan Timeline



Self-assessment

> Self-assessment is the process of reflecting on your own strengths, weaknesses, interests, and values. It is a way to learn more about yourself and to identify your goals and aspirations.

- ***Identify your interests***. What are you passionate about? What do you enjoy doing in your free time? What are your favorite courses you have taken?

- ***Identify your skills***. What are your strengths? What are you good at? What skills have you developed in your coursework and extracurricular activities?
  - *Don't worry as much about weaknesses– You are still learning, don't turn down career paths because you are 'not good at it' yet.*

- ***Identify your values***. What is important to you in a career? What kind of impact do you want to have on the world? What kind of work environment is important to you? What kind of work-life balance do you want?

Strengths & Weaknesses

- Analytical
- Creative
- Detail-oriented
- Efficient
- Empathetic
- Independent
- Willing to learn
- Innovative
- Leadership
- Organised
- Self-motivated
- Team player
- Versatile

- Disorganized
- Fear of failure
- Impatient
- Lack of experience
- Perfectionist
- Poor time management skills
- Unwilling to delegate tasks
- Procrastination
- Sensitive to criticism
- Shy
- Tendency to overwork themselves
- Uncomfortable with public speaking

Examples

- > Interest in using technology to solve social problems?
  - Social computing, social media, education technology, healthcare technology, environmental technology
- > Good at math and science, enjoys working on complex problems?
  - Software engineering, data science, machine learning, artificial intelligence, robotics
- > Creative and innovative?
  - Web design, UX design, game development, graphic design
- > Good at communicating and working with people?
  - Tech consultancy, sales, project management, teaching

Examples of CS degree careers

- Software Engineer
- Data Science
- Cybersecurity
- UX Design
- Business Analyst
- Technical writing (documentation, etc.)
- Sys Admin
- QA Engineer (software testing)
- Consultant
- Teacher
- Researcher
- Bioinformatics & Healthcare
- Engineer

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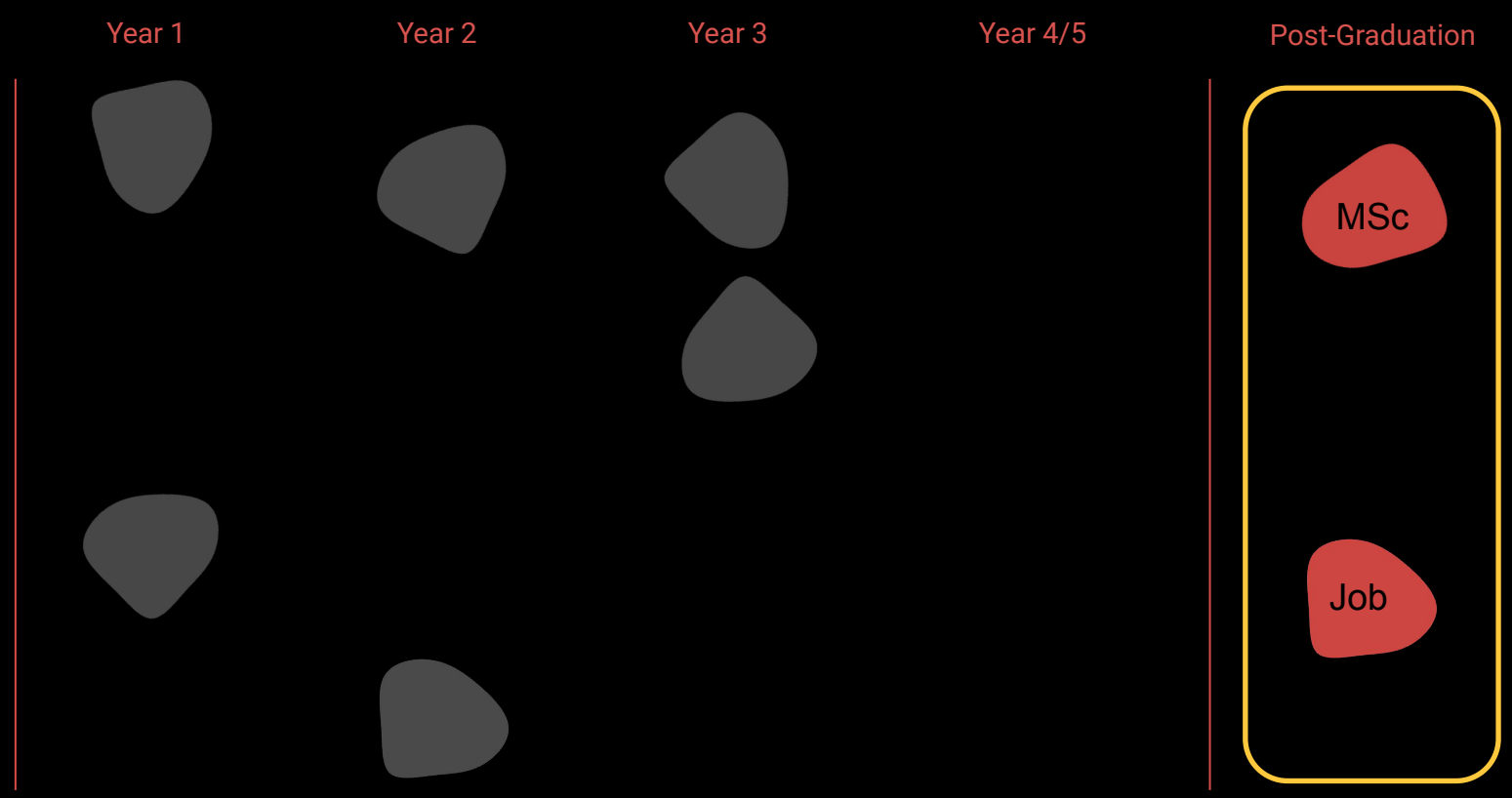

Honours Project

- During year 4 (and 5 for MInf) you will do a large project as part of a dissertation.
- Try and find if there is a lab doing something that you are interested in or links to your post-graduation plans
  - See ed.ac.uk/informatics/people/academic
- Read into the topic, later in year 3, you can discuss with your preferred academic and see if they would advise your UG4 project.
  - *InfPALS will explain more about this in Semester 2, so do not panic!*

How to find a lab

> Informatics is broken into multiple institutes

- ANC - Institute for Adaptive & Neural Computation
  - https://web.inf.ed.ac.uk/anc
- AIAI - Artificial Intelligence and its Applications Institute
  - https://web.inf.ed.ac.uk/aiai
- ILCC - Institute for Language, Cognition & Computation
  - https://web.inf.ed.ac.uk/ilcc
- ICSA - Institute for Computing Systems Architecture
  - https://web.inf.ed.ac.uk/icsa/
- IPAB - Institute of Perception, Action & Behaviour
  - https://web.inf.ed.ac.uk/ipab
- LFCS - Laboratory for Foundations of Computer Science
  - https://web.inf.ed.ac.uk/lfcs

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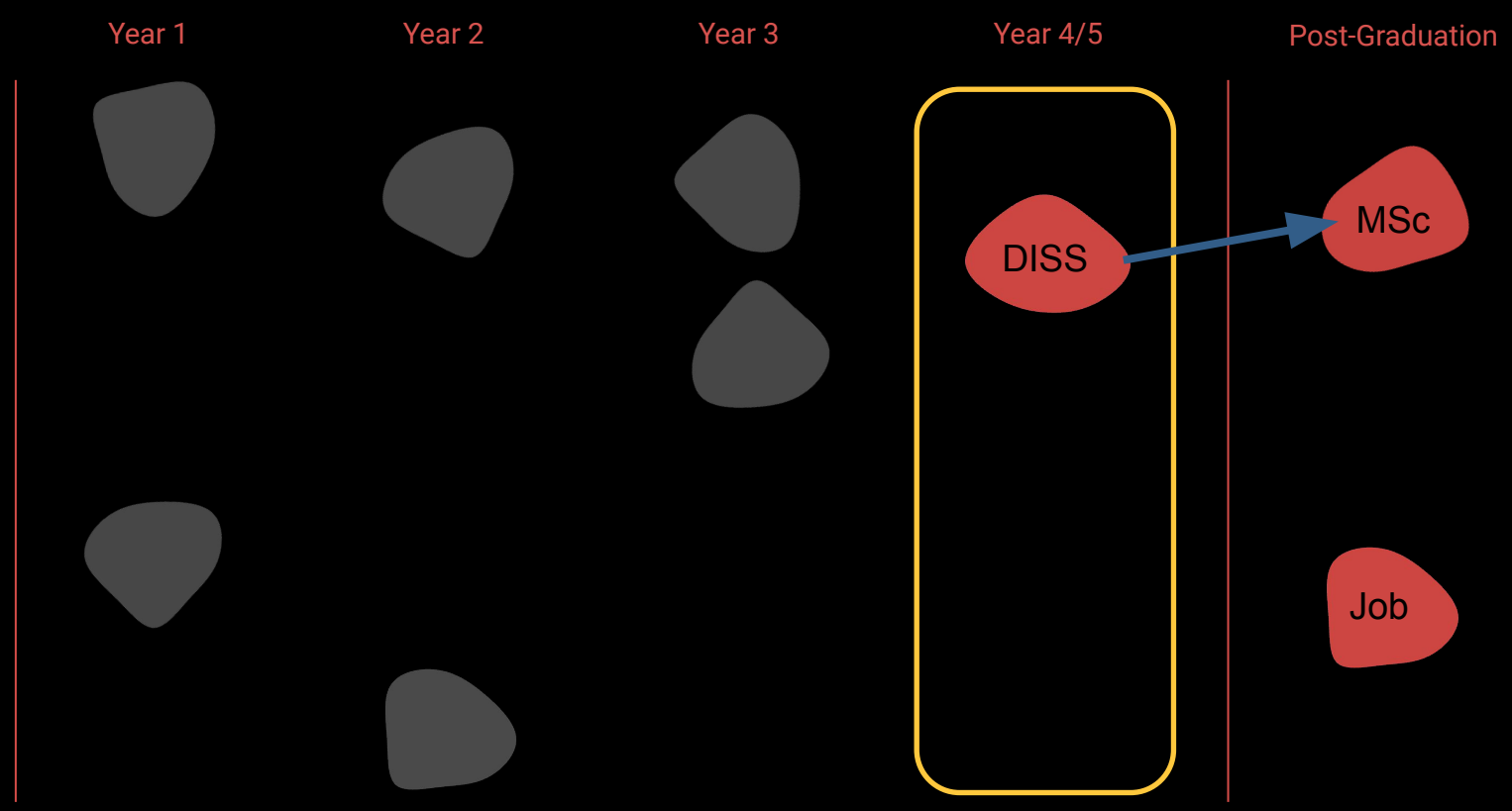
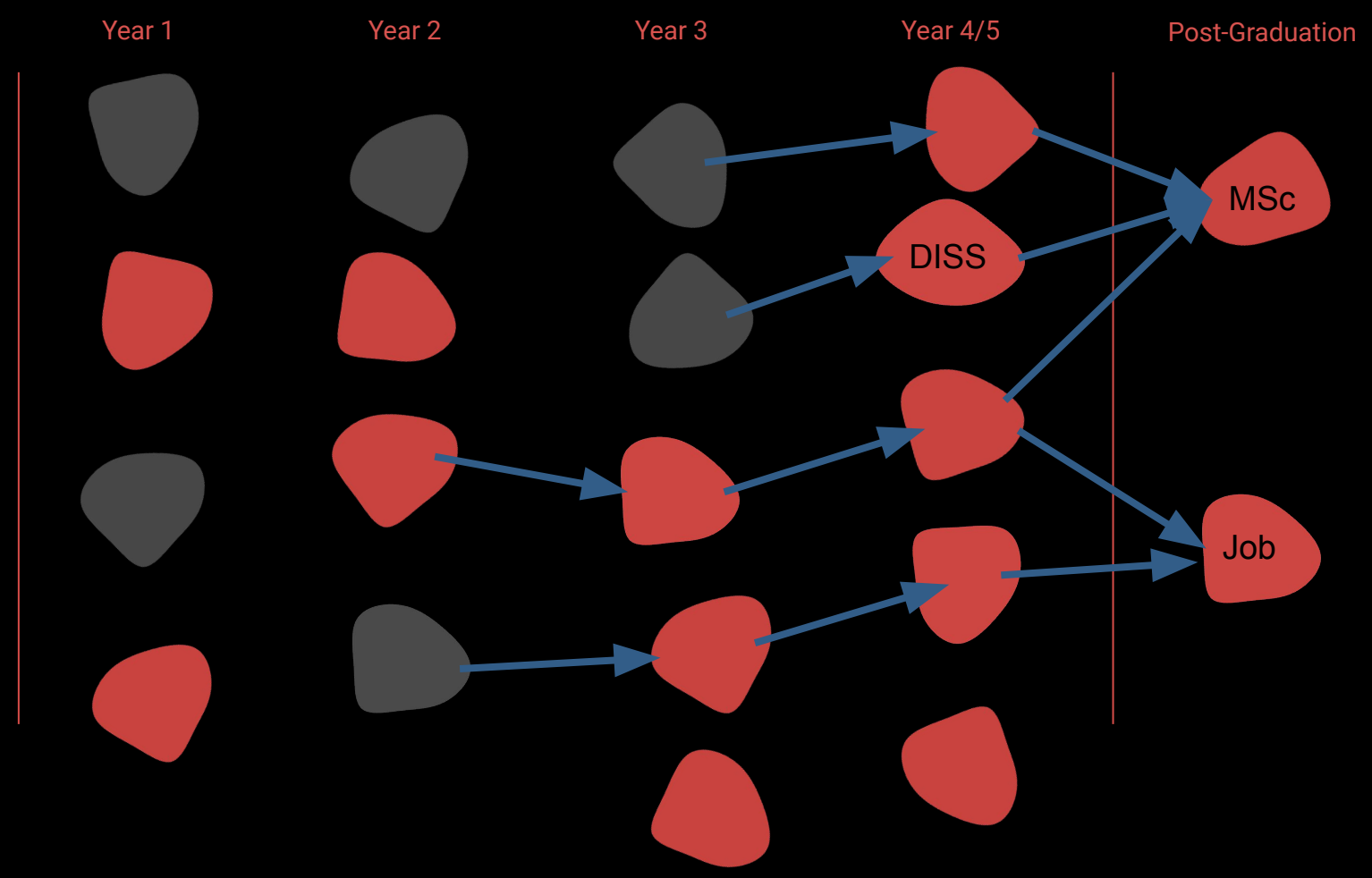


Choosing Courses

- Courses may change slightly or be removed entirely. Equally, new courses are introduced (e.g., new Entrepreneurship and Innovation course in year 3)
- Having a rough plan can still help with direction of your studies.
- Use PATH to plan into future years.
  - Look if those future courses have prerequisites.
- Consider outside courses to try new topics!

Career Plan Timeline



Experience

> **Why gain extracurricular experience?**
- Graduate jobs and PGT/PGR places are competitive.
- You may find you are interested in something that isn't taught at university (e.g., InfPALS skills workshops)

> **Gaining work experience can come in many forms**
- Internships
- Societies
- Roles within the university (e.g., year rep)
- Part-time jobs

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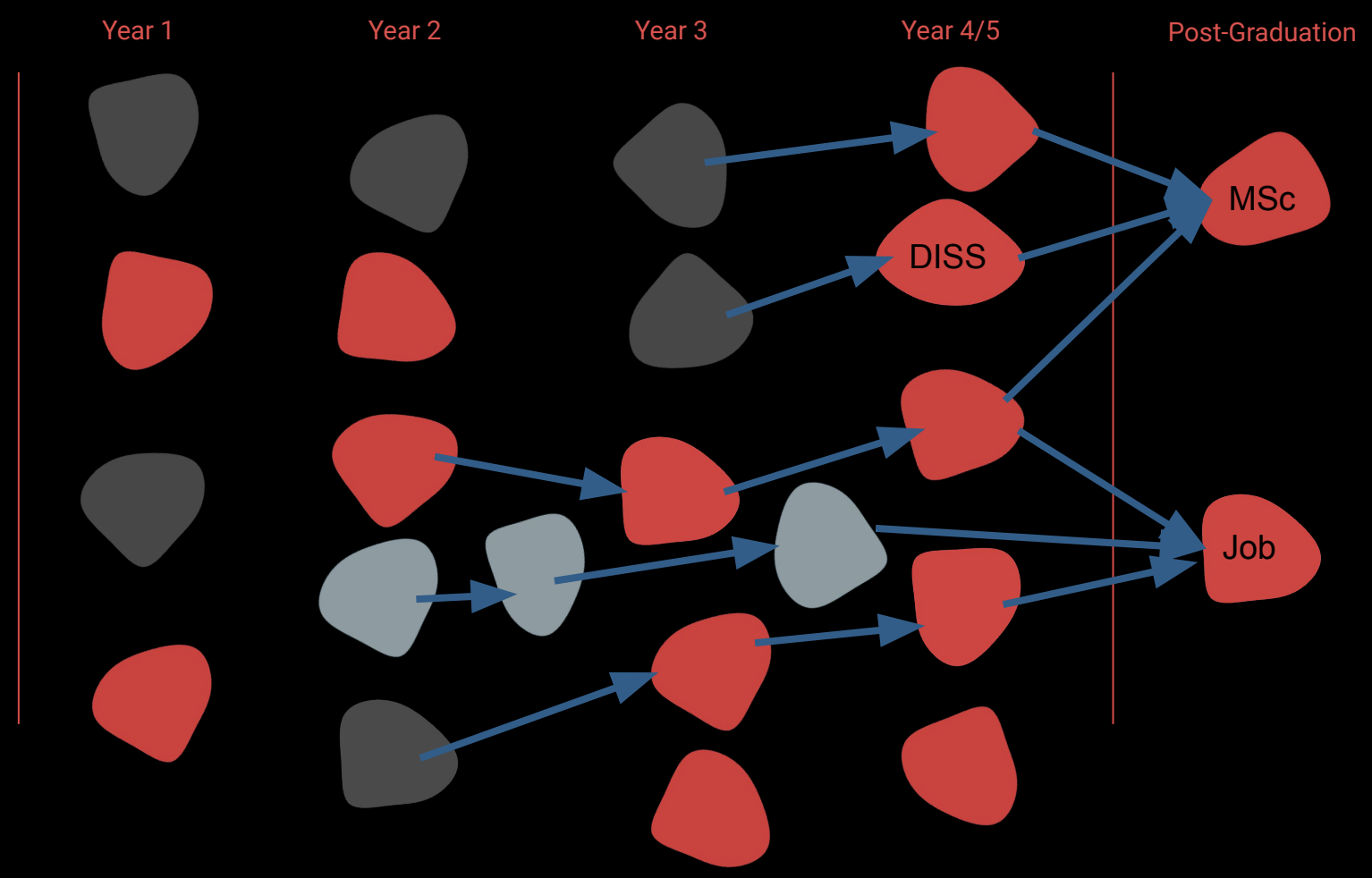

Your turn!

- Depending on how developed your career goals are already, use this time to continue developing a plan
  - Don't know what you want to do yet?
    -> Choosing a career interest
  - Know your area of interest?
    -> Use PATH to look into future courses
    -> Investigate labs/advisers for UG4/5 project
  - If you have done all the above
    -> Research work experience opportunities, or ways to develop outside the classroom
- Any questions, ask the leaders.
- InfPALS will be helping you with each of these areas in more depth next semester, so no rush to get everything perfect right away!